Chisom Anyabolu

Computer Science | Computational Biology | Visual Analytics | AI / Machine Learning

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RESEARCH PROFILE

Computer scientist with a PhD in Computer Science (magna cum laude) and research experience spanning computational biology, visual analytics, scientific software and the experimental evaluation of AI systems. Work includes genome annotation and RNA-sequencing analysis; interactive tools for DNA data storage and biological data interpretation; reproducible benchmarking; and uncertainty-aware human oversight. Current methodological research examines how scientific quantities derived from learned representations depend on the evidence-supported assumptions used to structure learning.

RESEARCH & TECHNICAL SKILLS

Research methods Experimental design, model evaluation, reproducible benchmarking, scientific visualization, human-AI interaction, task-based user evaluation, uncertainty communication

AI / machine learning PyTorch, TensorFlow, scikit-learn; neural networks, representation learning, PCA, t-SNE, grouped validation, repeated training and comparative analysis

Bioinformatics Genomics, RNA-seq, gene expression, alternative splicing, genome annotation, sequence analysis, scientific data integration

Programming & visualization Python, R, JavaScript, React, D3, Vega-Lite, Altair, Plotly, Matplotlib

Research computing Git, Docker, Jupyter, high-performance computing, SQL, NoSQL, LaTeX; computational pipelines and interactive research software

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RESEARCH EXPERIENCE

Postdoctoral Researcher

Jan 2024–Present

Robert Koch Institute (RKI), Center for Artificial Intelligence in Public Health Research, Visualization Group

- Developed ActionCue, a research framework and interactive visualization showing how multiple uncertainty conditions combine into oversight responses for AI-supported decisions, with worked examples in high-stakes settings.
- Designed and implemented controlled neural representation-learning experiments in PyTorch to investigate how scientific quantities depend on evidence-supported assumptions structuring the learning process. Compared full retraining with fixed-model estimates and limited adaptation across repeated experimental runs and optimization seeds.
- Designed and executed an antimicrobial-resistance genomics feasibility study of alternative scientific accounts for TEM-1-mediated co-amoxiclav resistance in Escherichia coli; implemented two learning conditions with tree-clade-grouped validation and repeated experimental runs.
- Collaborated on antimicrobial-susceptibility co-testing and membrane-protein annotation research, providing scientific review, interpretation of benchmarking results and manuscript development.
- Analysed and visualized findings from NLP research on German social-media discourse during COVID-19, and participated in conceptual research and scientific discussions on persona-adaptable language-model strategies.
- Develop independent research, lead and contribute to research proposals across computational and biomedical fields, and co-supervise doctoral research.

Research Associate

Oct 2020–Dec 2023

Heider Lab / MOSLA, Philipps-Universität Marburg, Germany

- Developed DNAsmart, an interactive tool for multi-attribute exploration and ranking of DNA data-storage sequences; demonstrated domain use cases and conducted an expert-user study.
- Co-designed UNACORM, a unified application for comparative evaluation of DNA data-storage codecs; helped design its user experience, validated experimental results, and supervised student-led codec reimplementations.
- Enhanced CAPT's user interface and developed its online version for interactive exploration of phylogenetic trees and taxonomic context.
- Conducted research on DNA-based data-storage design and evaluation, including analysis of encoding choices, error-related constraints and criteria for comparing storage approaches.
- Supervised and mentored Bachelor's and Master's students and supported teaching in bioinformatics, DNA data storage and data visualization.

Data Scientist / Research Supervisor

May 2018–Oct 2019

International Institute of Tropical Agriculture (IITA), Ibadan, Nigeria

- Developed Python and R workflows for RNA-sequencing analysis to investigate gene expression and alternative splicing in biological datasets.
- Processed large sequencing datasets using high-performance computing infrastructure, integrated analytical outputs and performed data-quality checks to support interpretable scientific results.
- Prepared research reports, figures and presentations, and supported analysis coordination and research supervision within computational genomics projects.

Research Fellow

May 2017–Apr 2018

International Institute of Tropical Agriculture (IITA), Ibadan, Nigeria

- Developed GeAnnotate, a computational pipeline for improving cassava genome annotation where existing gene annotations were incomplete; implemented supporting genomic processing and analysis in Python and R.
- Contributed genomic analyses to research manuscripts, scientific presentations and project development.

EDUCATION

PhD (Dr. rer. nat.) in Computer Science | 2024 · Philipps-Universität Marburg, Germany

Magna cum laude. Dissertation: Visual Analytics Approaches to Genomic Data Representation: From Molecular Storage to Multidimensional Projection.

MSc in Computer Science (Bioinformatics) | 2018 · University of Ibadan, Nigeria

BSc in Computer Science | 2013 · Anambra State University, Nigeria

SELECTED RESEARCH SOFTWARE

ActionCue — Process-transparency visualization linking multiple uncertainty conditions to human oversight actions.

DNAsmart — Interactive multi-attribute ranking and sequence-exploration tool for DNA data storage; developed in React/JavaScript and evaluated with domain experts.

UNACORM — Unified application for standardized comparison and reproducible evaluation of DNA data-storage codecs; application co-design, UX and comparative validation.

CAPT — Context-aware phylogenetic and taxonomic visualization; interface and online implementation.

GeAnnotate — Cassava genome-annotation pipeline developed at IITA to address incomplete gene annotations.

PUBLICATIONS

Earlier publications appeared under the name Chisom Ezekannagha.

Anyabolu, C., Dubey, A., & Hattab, G. (2026). Visualizing Uncertainty-to-Action Composition for Human Oversight. *IEEE VIS 2026 Uncertainty Visualization Workshop*. arXiv:2608.16428.

Anžel, A., **Anyabolu, C.**, Wimbes, L., Staus, L., Heldian, I. T., Sonnabend, D., Elhadri, K., Becker, S., Klein, F., Schoen, R., Schwarz, M., Welzel, M., Freisleben, B., Heider, D., & Hattab, G. (2026). The Unified Evaluation App for DNA Data Storage Codecs. *Manuscript under review*. arXiv:2608.09673.

Awotoro, E., Anžel, A., **Anyabolu, C.**, Willrich, N., & Hattab, G. (2026). Network-based characterization of latent co-testing patterns in antimicrobial susceptibility testing. *iScience*, 29(10), 117421. doi:10.1016/j.isci.2026.117421.

Awotoro, E. O., **Anyabolu, C.**, Schwarz, F., Tauscher, J., Heider, D., Ladewig, K., Le Bon, C., Moncoq, K., Miroux, B., & Hattab, G. (2026). MetaMP Ecosystem for Unified, Auditable, and Benchmark-Ready Data for Reliable Membrane Protein Annotation. *Computational and Structural Biotechnology Journal*, 35, 0165. doi:10.34133/csbj.0165.

Hattab, G., Anžel, A., Dubey, A., **Ezekannagha, C.**, Yang, Z., & İlgen, B. (2024). Persona Adaptable Strategies Make Large Language Models Tractable. *Proceedings of the International Conference on Natural Language Processing and Information Retrieval (NLPIR)*. doi:10.1145/3711542.3711581.

Wehrli, S., **Ezekannagha, C.**, Hattab, G., Boender, T. S., Arnrich, B., & Irrgang, C. (2024). Guiding Sentiment Analysis with Hierarchical Text Clustering: Analyzing the German X/Twitter Discourse on Face Masks in the 2020 COVID-19 Pandemic. *Proceedings of the 14th Workshop on Computational Approaches to Subjectivity, Sentiment, & Social Media Analysis (WASSA)*, pp. 153–167. doi:10.18653/v1/2024.wassa-1.13.

Ezekannagha, C., Welzel, M., Heider, D., & Hattab, G. (2023). DNAsmart: Multiple Attribute Ranking Tool for DNA Data Storage Systems. *Computational and Structural Biotechnology Journal*, 21, 1448–1460. doi:10.1016/j.csbj.2023.02.016.

Ezekannagha, C., Becker, A., Heider, D., & Hattab, G. (2022). Design Considerations for Advancing Data Storage with Synthetic DNA for Long-Term Archiving. *Materials Today Bio*, 15, 100306. doi:10.1016/j.mtbio.2022.100306.

Kaya, G., **Ezekannagha, C.**, Heider, D., & Hattab, G. (2022). Context-Aware Phylogenetic Trees for Phylogeny-Based Taxonomy Visualization. *Frontiers in Genetics*, 13, 891240. doi:10.3389/fgene.2022.891240.

Olagunju, T. A., **Ezekannagha, C.**, & Gisel, A. (2021). 3′-Tag RNA-Sequencing. *EMBnet.journal*, 26(Suppl. A), e968 (conference abstract). doi:10.14806/ej.26.A.968.

RESEARCH SUPERVISION & MENTORING

Doctoral research Ebenezer Awotoro, PhD in Computer Science (research co-supervision; expected 2027).

Master's research Florian Schwarz (MSc Computer Science, 2023); Minh Nguyen (MSc Business Informatics, 2023).

Bachelor's research Chu Xiao (BSc Computer Science, 2023).

Research mentoring Mentored student researchers and coordinated technical contributions within interdisciplinary computational research projects.

TEACHING

Introduction to Bioinformatics — Philipps-Universität Marburg, tutor (2022).

DNA Storage Encodings group project — Philipps-Universität Marburg, tutor and co-organizer (2021–2022).

Data Visualization — Philipps-Universität Marburg, co-tutor (2021).

Biological Data Visualization — Philipps-Universität Marburg, co-tutor (2020–2022).

PROFESSIONAL SERVICE, PRESENTATIONS & ENGAGEMENT

Programme Committee, IEEE VIS 2026 Uncertainty Visualization Workshop; scientific journal peer reviewing.

Panelist, Women Leading Innovation in Health & AI (Doctolib / Women Techmakers Berlin).

Selected conference presentations: International Conference on Data Storage in Molecular Media (Bielefeld, 2022); Nigerian Bioinformatics Conference (Lagos, poster, 2019).

TRAINING & GRANTS

Oxford Machine Learning Summer School, University of Oxford (2022).

Bio+Med+Vis Spring School (2021).

Next Generation Sequencing Bioinformatics Workshop, University of the Witwatersrand (2019; travel grant).

H3ABioNet Introduction to Bioinformatics Course, University of Ibadan (2016; training grant).

LANGUAGES & CAREER LEAVE

English — fluent **Igbo** — native **German** — B1

Parental leave: 2025–2026.